

Lab 07 Template

- 1.) Screenshot of properly configured DNS MX query
(5 points)

```
cpre230@ubuntuserver:~$ dig MX student11.230.com @199.100.16.100

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>> MX student11.230.com @199.100.16.100
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12186
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 1ef4acc3e1086cd70100000067e03b9fe911cce98096e905 (good)
;; QUESTION SECTION:
;student11.230.com.                IN      MX

;; ANSWER SECTION:
student11.230.com.                58      IN      MX      10 mail.student11.230.com.

;; ADDITIONAL SECTION:
mail.student11.230.com. 659     IN      A      42.49.30.204

;; Query time: 0 msec
;; SERVER: 199.100.16.100#53(199.100.16.100) (UDP)
;; WHEN: Sun Mar 23 16:49:35 UTC 2025
;; MSG SIZE rcvd: 111

cpre230@ubuntuserver:~$ _
```

2.) Screenshot of output from netstat -tnl
(5 points)

```
The following NEW packages will be installed:
net-tools
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 204 kB of archives.
After this operation, 819 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 net-tools amd64 1.60+git20181103.0eebece-1ubuntu5 [204 kB]
Fetched 204 kB in 0s (1,489 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 146644 files and directories currently installed.)
Preparing to unpack .../net-tools_1.60+git20181103.0eebece-1ubuntu5_amd64.deb ...
Unpacking net-tools (1.60+git20181103.0eebece-1ubuntu5) ...
Setting up net-tools (1.60+git20181103.0eebece-1ubuntu5) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
cpre230@mail:~$ netstat -tnl
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:995             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:993             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:110             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:143             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:53              0.0.0.0:*               LISTEN
cpre230@mail:~$ _
```

- 3.) Screenshot of Sending mail from cpre230a --> cpre230b in terminal
(10 points)

```
cpre230@mail:~$ telnet mail.student11.230.com 25
Trying 192.168.1.204...
Connected to mail.student11.230.com.
Escape character is '^]'.
220 mail.home.arpa ESMTP Postfix (Ubuntu)
helo student11.230.com
250 mail.home.arpa
mail from: <cpre230a@student11.230.com>
250 2.1.0 Ok
rcpt to: <cpre230b@student11.230.com>
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Subject: Spring break
How was your spring break? I hope it was nice.
.
250 2.0.0 Ok: queued as 8EDA8D070
quit
221 2.0.0 Bye
Connection closed by foreign host.
cpre230@mail:~$ _
```

- 4.) Screenshot of reading the previously sent message in terminal
(10 points)

```
cpre230@mail:~$ telnet mail.student11.230.com 110
Trying 192.168.1.204...
Connected to mail.student11.230.com.
Escape character is '^]'.
+OK Dovecot (Ubuntu) ready.
USER cpre230b
+OK
PASS cpre230
+OK Logged in.
LIST
+OK 1 messages:
1 539
.
RETR 1
+OK 539 octets
Return-Path: <cpre230a@student11.230.com>
X-Original-To: cpre230b@student11.230.com
Delivered-To: cpre230b@student11.230.com
Received: from student11.230.com (mail.student11.230.com [192.168.1.204])
        by mail.home.arpa (Postfix) with SMTP id 8EDA8D070
        for <cpre230b@student11.230.com>; Sun, 23 Mar 2025 21:37:07 +0000 (UTC)
Subject: Spring break
Message-Id: <20250323213744.8EDA8D070@mail.home.arpa>
Date: Sun, 23 Mar 2025 21:37:07 +0000 (UTC)
From: cpre230a@student11.230.com

How was your spring break? I hope it was nice.

.
QUIT
+OK Logging out.
Connection closed by foreign host.
cpre230@mail:~$ _
```

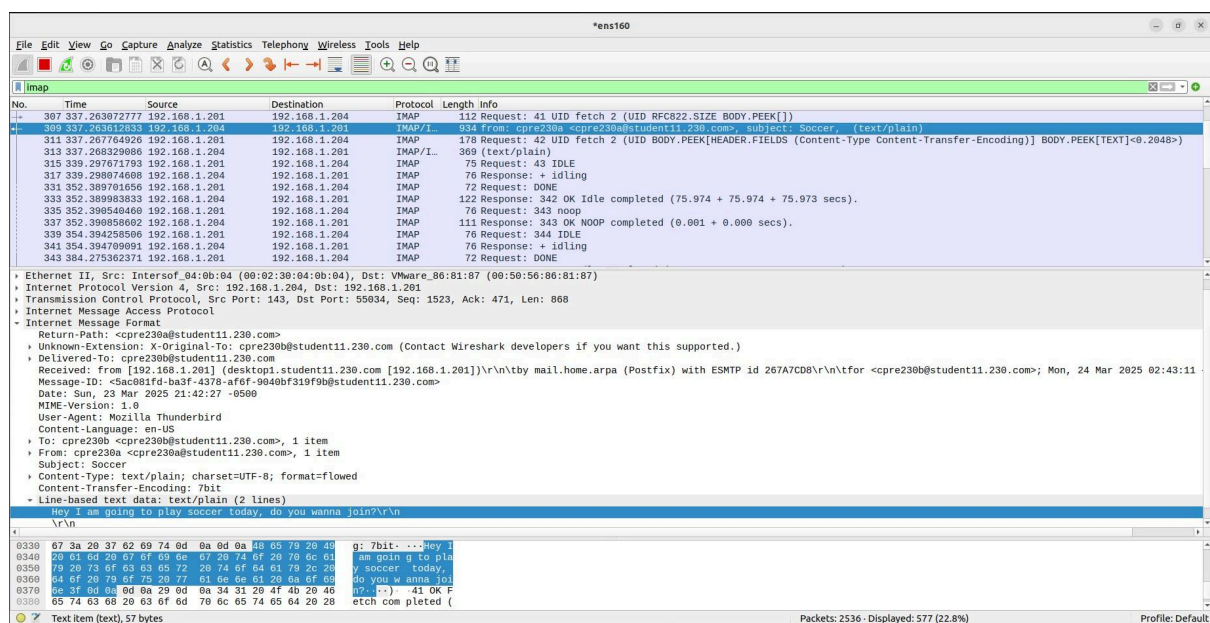
5.) Screenshot of Dovecot cert raw contents
(5 points)

[illegible]

6.) Screenshot of Dovecot cert parsed as X.509
(5 points)

```
Certificate:
Data:
  Version: 3 (0x2)
  Serial Number:
    2c:fa:04:4a:40:9f:3f:59:bb:a0:1d:2e:00:58:82:37:0d:98:4d:83
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: O = Dovecot mail server, OU = mail.student11.230.com, CN = mail.student11.230.com, emailAddress = postmaster@student11.230.com
  Validity
    Not Before: Mar 23 21:52:48 2025 GMT
    Not After : Mar 23 21:52:48 2026 GMT
  Subject: O = Dovecot mail server, OU = mail.student11.230.com, CN = mail.student11.230.com, emailAddress = postmaster@student11.230.com
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:c2:12:55:93:fe:b2:ae:63:b1:ac:f4:e5:a6:bd:
      42:74:3d:77:6d:6b:70:3e:dd:63:4a:a7:56:80:31:
      2f:41:66:8f:10:2d:ca:dc:b2:7c:89:8c:82:9c:66:
      38:21:f7:ae:1c:89:eb:da:16:3b:f5:f6:27:8f:59:
      2d:81:a7:bc:66:fc:cf:a1:d1:59:44:41:98:13:81:
      68:0e:03:53:c6:fb:91:8e:0f:b1:3c:41:2f:9c:5b:
      bd:b3:2d:60:6b:30:18:04:61:84:d2:a7:30:71:f2:
      b8:2a:d2:f4:3d:5f:87:70:6d:8d:5b:8e:94:dc:f6:
      7e:42:82:2c:e9:31:56:e0:7e:47:02:2b:01:5e:83:
      f1:c3:7e:6a:16:5b:c4:11:04:88:21:3c:e3:12:3e:
      b0:65:43:51:1e:4e:90:11:39:3b:30:e4:ce:62:3b:
      3c:54:0d:3b:ba:51:cc:a5:89:f3:bc:f5:b1:b8:12:
      52:fa:ac:3b:b4:d1:93:54:1c:9c:0d:72:12:70:78:
      15:0d:10:22:e2:3b:63:5c:d4:f1:b5:2f:78:0a:86:
      60:82:a2:d7:cb:86:ce:7a:35:00:36:11:33:f2:97:
      f2:62:13:6a:54:97:f4:ca:22:0e:f2:68:68:5f:16:
      9d:a9:eb:6f:51:ef:63:40:54:2e:b6:69:da:58:e9:
      f0:cb
    Exponent: 65537 (0x10001)
```

7.) Screenshot of plaintext IMAP - Wireshark
(5 points)



8.) Screenshot of Base64 decode of password (10 points)

The screenshot shows a web browser window with the URL `https://emn178.github.io/online-tools/base64_decode.html`. The page title is "Base64 Decode". Below the title, a description states: "This online Base64 decoding tool helps you decode Base64 to text or binary. You can output UTF-8, UTF-16, Hex, or other encodings. It also supports various formats such as RFC 4648 (standard and URL Safe), RFC 2045 (MIME), RFC 2152 (UTF-7), and RFC 3501 (IMAP)."

The interface is divided into two main sections: "Settings" on the left and "Input/Output" on the right.

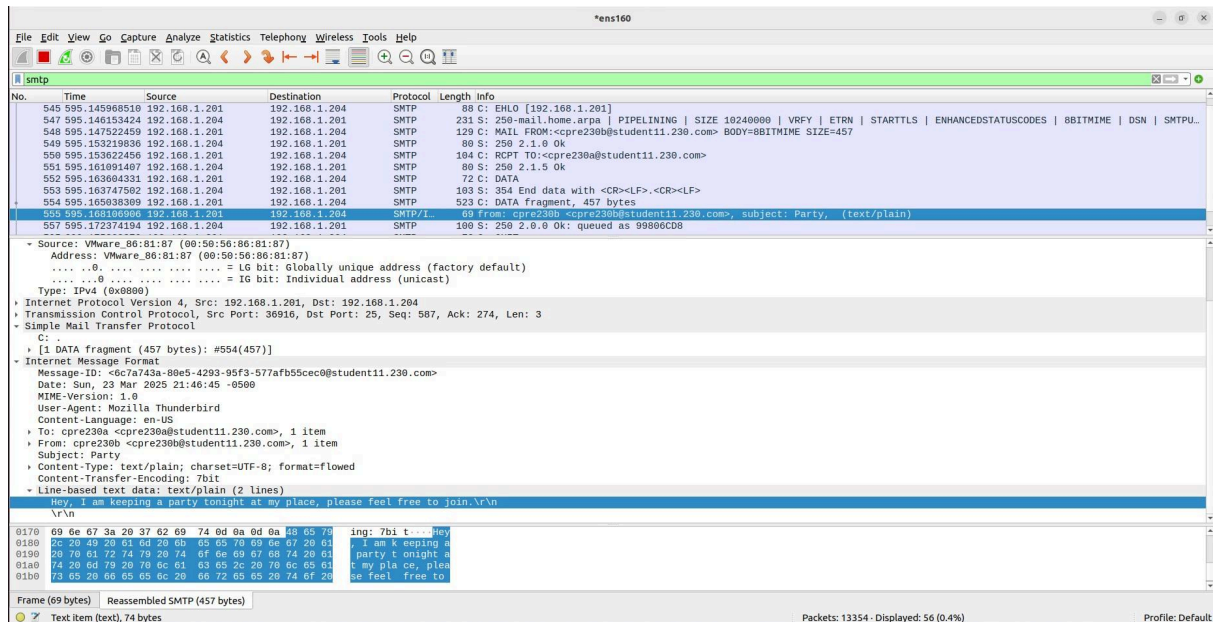
Settings:

- A "Decode" button is at the top.
- There are two toggle switches: "Auto Update" (checked) and "Remember Input" (unchecked).
- Under "Output Encoding", a dropdown menu is set to "UTF-8".

Input/Output:

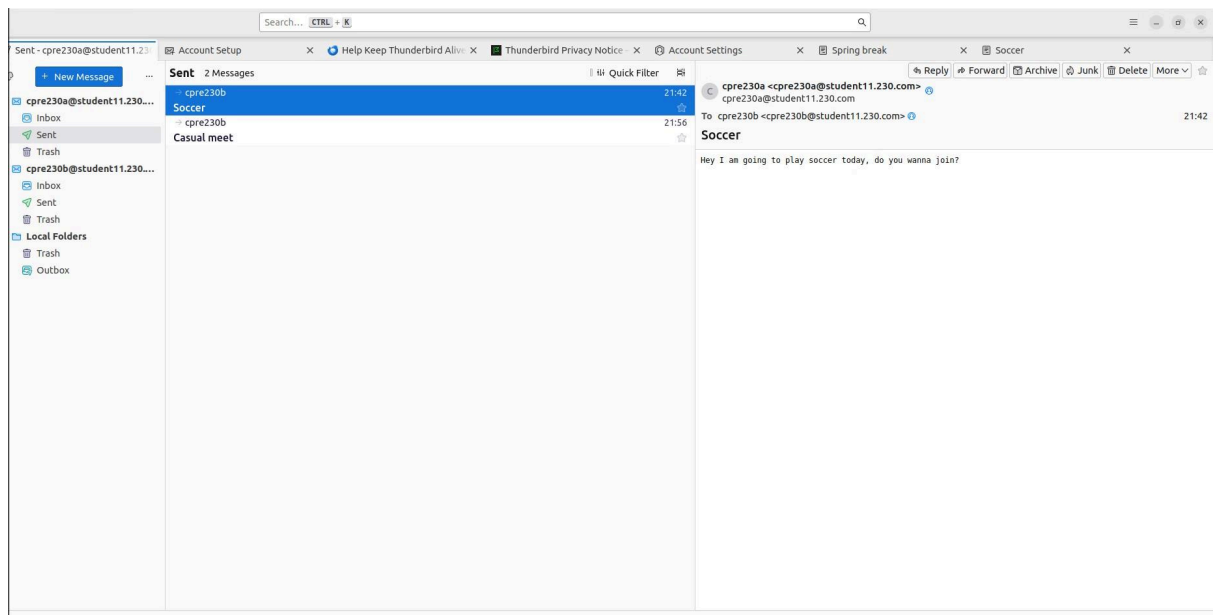
- The "Input" field contains the Base64 string: `AGNwcmUyMzBhAGNwcmUyMzA=`.
- The "Output" field displays the decoded result: `□cpre230a□cpre230`.
- At the bottom, there is a "Share Link" section with a copy icon.

9.) Screenshot of plaintext SMTP - Wireshark (5 points)

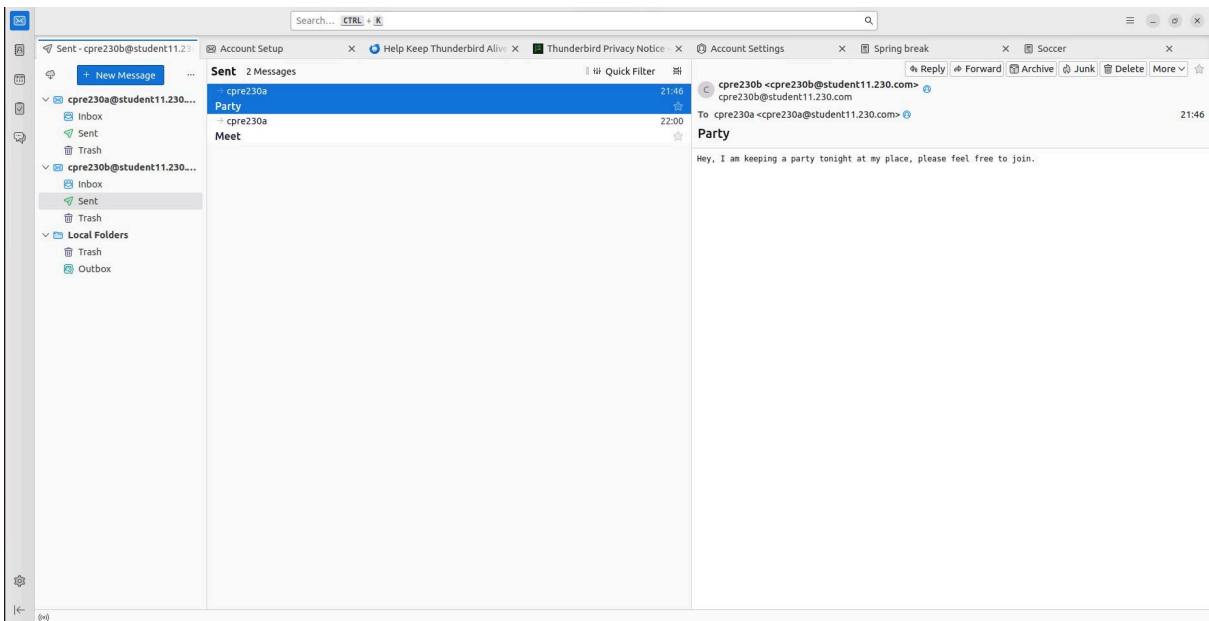


10.) Screenshot of test messages in Thunderbird (5 points)

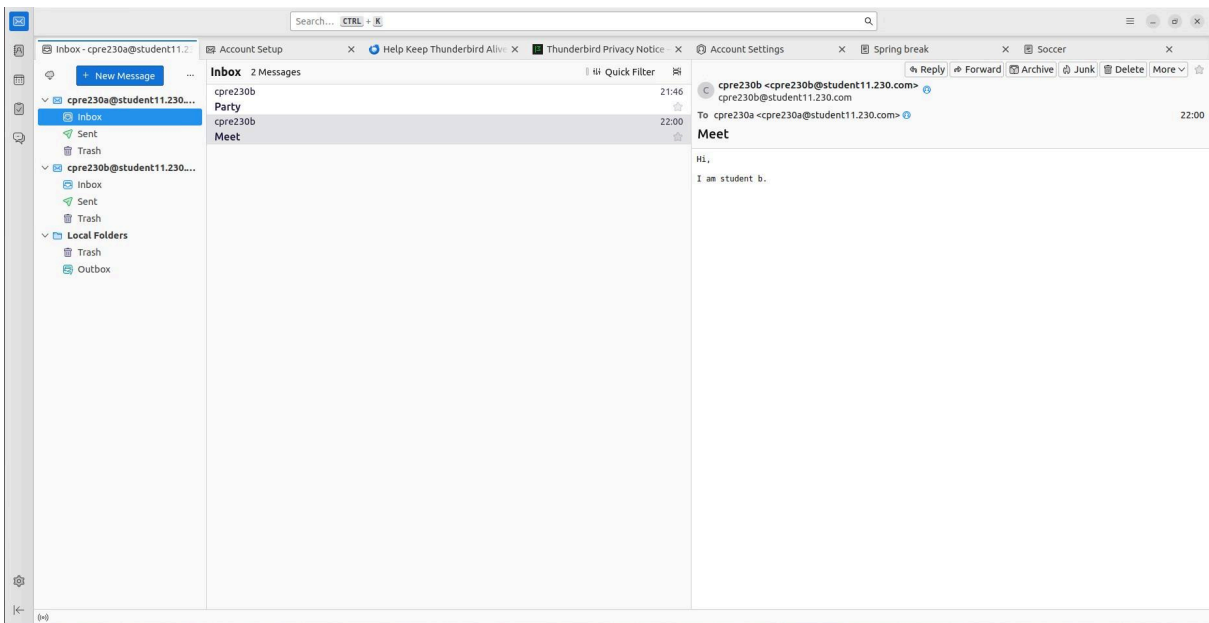
Messages sent by user a



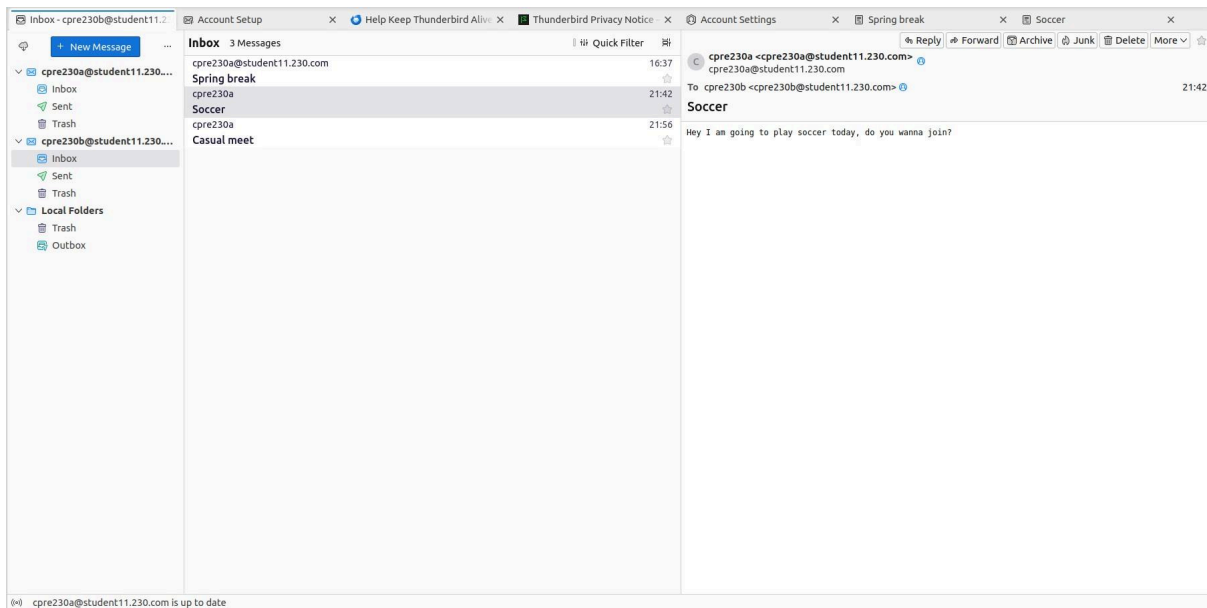
Messages sent by user b



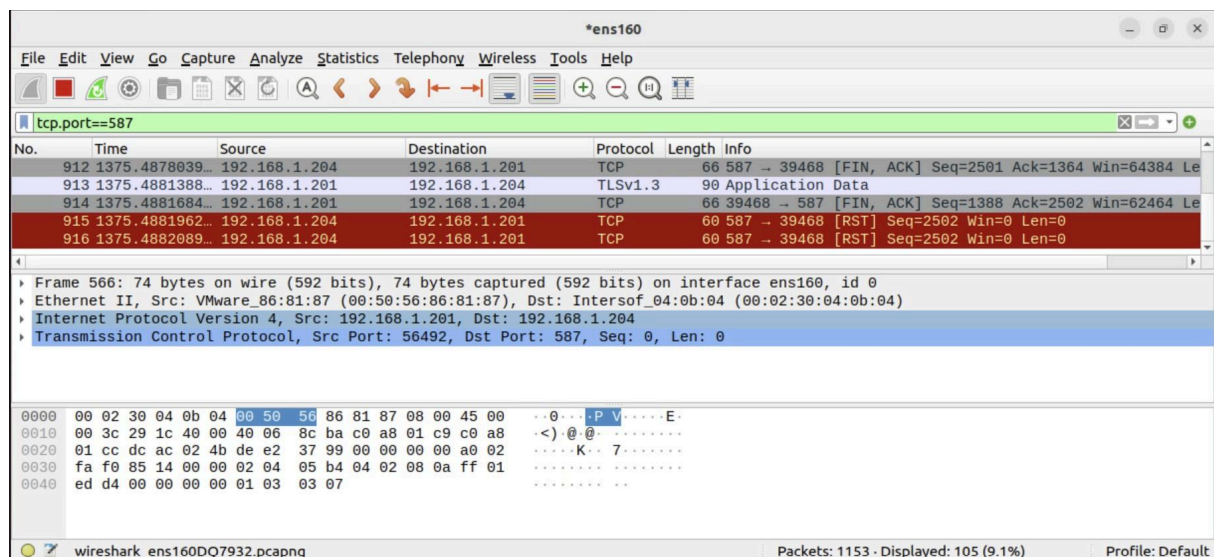
Messages received by user a



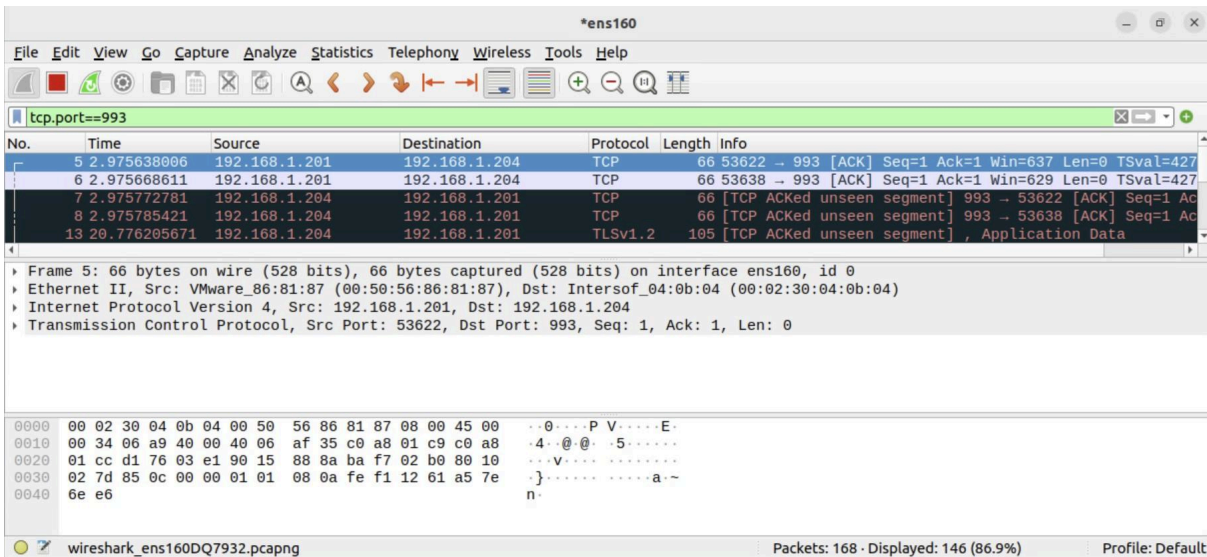
Messages received by user b



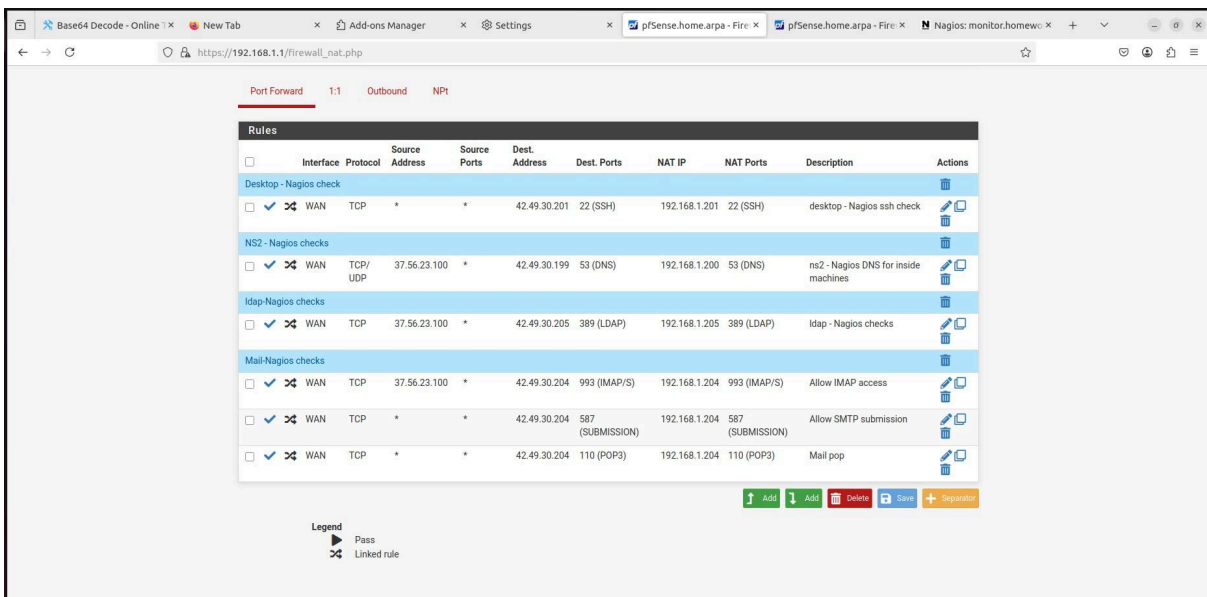
11.) Screenshot of encrypted SMTP - Wireshark (5 points)



12.) Screenshot of encrypted IMAP - Wireshark
(5 points)



13.) Screenshot of NAT rules
(5 points)



14.) Screenshot of Nagios queries with green indicated for all mail services
(20 points)

Student 11 (student11)

Host	Services	Actions
desktop1.student11.230.com	desktop1 ssh	  
ldap.student11.230.com	ldap Bezos ldap Gates ldap Jobs ldap Musk ldap Zuckerberg ldap service	  
mail.student11.230.com	dns MX mail pop secure imap secure imap login secure smtp	  
ns1.student11.230.com	dns mail dns ns1 dns www dns www2 reverse dns mail reverse dns ns1 reverse dns www reverse dns www2	  
ns2.student11.230.com	dns desktop1 dns ldap dns ws reverse dns desktop1 reverse dns ldap reverse dns ws	  
www.student11.230.com	www http www https www login	  
www2.student11.230.com	www2 django www2 ssh	  